Quiz-2

Que-1)

$x[n] = d(n) + 2d(n-1) + 2d(n+1)$
$h[n] = 5d(n) - 6d[n-2] + 9d(n+1)$

a) convolution

$y[n] = x[n] * h[n]$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

$x[-1] = 2$ $\quad$ $h[-1] = 9$
$x[0] = 1$ $\quad$ $h[0] = 5$
$x[1] = 2$ $\quad$ $h[2] = -6$
$\quad$ $h[1] = 0$

n	y[n]
-2	$2(9) = 18$
-1	$2(5) + (1)(9) = 10 + 9 = 19$
0	$2(0) + 1(5) + 2(9) = 23$
1	$1(0) + 2(5) = 10$
2	$2(0) = 0$

$y[n] = \{18, 19, 23, 10, 0\}$

b) circular convolution

$N = 3$

$$y_c[n] = \sum_{k=0}^{N-1} x[k] \cdot h[(n-k) \bmod N]$$

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$n = 0, 1, 2$

$y_c[0] = 1*5 + 2(-6) + 2(0) = -7$
$y_c[1] = 1(0) + 2(5) + 2(-6) = -2$
$y_c[2] = 1(-6) + 2(0) + 2(5) = 4$

$$y_c[n] = \{-7, -2, 4\}$$

c) cross correlation

$$r_{xh}[k] = \sum_n x[n] \cdot h[n+k]$$

$k = -2, -1, 0, 1, 2$

k	$r_{xh}[k]$
-2	$0 + 0 + 2(9) = 18$
-1	$0 + 1(9) + 2(5) = 19$
0	$2(9) + 1(5) + 2(0) = 23$
1	$2(5) + 1(0) + 2(-6) = -2$
2	$0 + 1(-6) + 0 = -6$

$$r_{xh}[k] = \{18, 19, 23, -2, -6\}$$

d) Autocorrelation of $x[n]$

$$r_{xx}[k] = \sum_n x[n] x[n+k]$$

$k = -2, -1, 0, 1, 2$

$k = -2 \quad = -1(22) = 4$
$k = -1 \quad 21(12) = 4$
$k = 0 \quad x(-1)^2 + x(0)^2 + x(1)^2 = 4 + 1 + 4 = 9$
$k = 1 \quad x(0) + x[-1] + x[1]x(0) = 12*21 = 4$

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$$\gamma_{xx}[k] = \{4, 4, 9, 4, 4\}$$

Question-2)

mput image:

$$\begin{matrix} 1 & 5 & 7 & 8 \\ 9 & 3 & 2 & 6 \\ 1 & 2 & 3 & 9 \end{matrix}$$

Kernal

$$\begin{matrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{matrix}$$

zero padding

$$I_{pad} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 5 & 7 & 8 & 0 \\ 0 & 9 & 3 & 2 & 6 & 0 \\ 0 & 1 & 2 & 3 & 9 & 0 \end{bmatrix}$$

Filtered Image

$$\begin{bmatrix} 0+0+0+0+1\ 1+3=4 & 1(0)+0(0)+1(0)+5(1)+9(0)=16 & 16 & 10 \\ 1(0)+9(0)+9(3)+2(3)=16 & 3(1)+4(3)+1(0)+5(1)+1(1)=15 & 26 & 16 \\ 1(1)+4(0)+3(0)+3(1)=4 & 4(1)+1(2)(3)+4(5)+2(3) & 12 & 11 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 16 & 16 & 10 \\ 16 & 15 & 26 & 16 \\ 4 & 13 & 12 & 11 \end{bmatrix}$$

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Question -3)

input image :
$$\begin{matrix} 1 & 5 & 7 & 8 \\ 9 & 3 & 2 & 6 \\ 1 & 2 & 3 & 9 \end{matrix}$$

Zero Padding :
$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 5 & 7 & 8 & 0 \\ 0 & 9 & 3 & 2 & 6 & 0 \\ 0 & 1 & 2 & 3 & 9 & 0 \end{bmatrix}$$

a) smoothening filter

$$I(i,k) = \frac{1}{9} \sum_{3\times 3} \text{pixels}$$

Smoothened Image

$$I_{avg} = \begin{bmatrix} 18/9 & 27/9 & 31/9 & 23/9 \\ 21/9 & 33/9 & 45/9 & 35/9 \\ 15/9 & 20/9 & 25/9 & 20/9 \end{bmatrix}$$

$$SI = \begin{bmatrix} 2 & 3 & 3 & 3 \\ 2 & 4 & 5 & 4 \\ 2 & 2 & 3 & 2 \end{bmatrix}$$

b) median filter

$$\begin{bmatrix} 0,0,0,0,1,5,0,9,3 & 3 & 3 & 0,0,0,1,5,7,9,3,2 \\ 0,0,0,5,7,8,3,2,6 & 3 & 6 & 0,0,0,7,8,0,2,6,0 \\ 1 & 2 & 3 & 2 \end{bmatrix}$$

$$I_{median} = \begin{bmatrix} 0 & 3 & 3 & 0 \\ 3 & 3 & 6 & 3 \\ 1 & 2 & 3 & 2 \end{bmatrix}$$

c) max filter

$$\begin{bmatrix} 0,0,0,1,5,0,9,3 & & & \\ 9 & 9 & 9 & 9 \\ 9 & 9 & 9 & 9 \end{bmatrix}$$

$$\begin{bmatrix} \text{max} & & & \\ 0,0,0,0,1,5,0,9,3 & 0,0,0,1,5,7,9,3,2 & 0,0,0,5,7,8,3,2,6 & 0,0,7,8,0,2,6 \\ 9 & 9 & 9 & 9 \\ 9 & 9 & 9 & 9 \end{bmatrix}$$

$$I_{max} = \begin{bmatrix} 9 & 9 & 8 & 8 \\ 9 & 9 & 9 & 9 \\ 9 & 9 & 9 & 9 \end{bmatrix}$$

Question-4)

Input image $f(x,y) = \begin{bmatrix} 1 & 5 & 7 \\ 9 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$

a) df/dx

$$K_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

zero Padding

$$F_{pad} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 5 & 7 & 0 \\ 0 & 9 & 3 & 2 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$df/dx = \begin{bmatrix} 8 & -1 & -8 \\ -7 & -4 & 3 \\ 1 & 3 & 3 \end{bmatrix}$$

b) df/dy (vertical derivates)

$$K_y = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$df/dy = \begin{bmatrix} 12 & 14 & 5 \\ -2 & -2 & 3 \\ -10 & -10 & -9 \end{bmatrix}$$

c) laplacian

$$K_L = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\text{laplacian} = \begin{bmatrix} -6 & -12 & -10 \\ 9 & 6 & -2 \\ -11 & -2 & 8 \end{bmatrix}$$